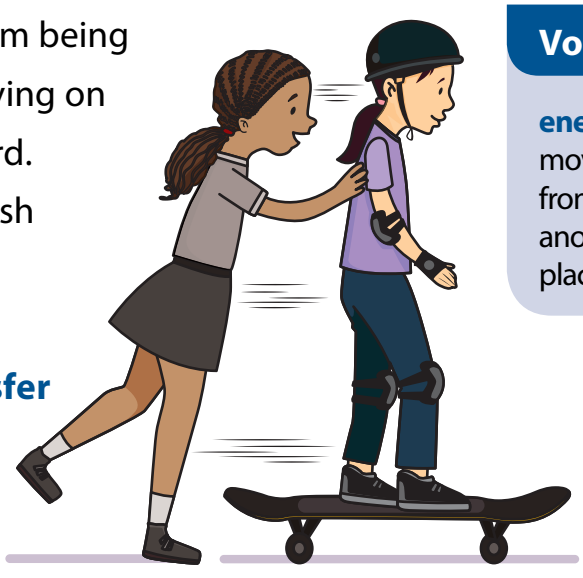
Energy Transfer

Chapter 4

Your friend is standing still on a skateboard. She asks you to give her a gentle push. Your friend starts out at rest, but when you push on her back she moves down the sidewalk, rolling away from you.

Energy causes that change. When you push your friend, your motion energy causes her to change from being at rest to moving on the skateboard. When you push your friend, you cause an **energy transfer** from you to your friend.



Big Question

What evidence shows that energy transfers from place to place?

Vocabulary

energy transfer, n. movement of energy from one object to another or from one place to another

Word to Know

Evidence is anything that helps prove or disprove an idea. Your own observations or information and facts presented by others can be used as evidence. Scientists and engineers use evidence to support explanations about how and why things happen and to improve their solutions to problems.

Energy Transfers Between Objects in Contact

If you kick a ball so that it hits a second ball, what happens? When you kick the first ball, you transfer energy from your body to the first ball. The position of the ball changes as it moves.

When the first ball hits the second ball, it transfers its motion energy to the second ball. Now, the position of the second ball changes, too. These changes in motion are evidence that energy has transferred from one object to another.



Motion energy transfers from the kicker's foot to the ball, causing a change of the ball's position.

Energy Transfers in Different Ways

All forms of energy—including light, heat, electricity, and sound—are ways that energy transfers, or moves. Energy can transfer from one object to another. Energy can also transfer from one place to another. You have seen examples of energy transfer between objects that touch each other. But are you in contact with the sun when you see its light? Light can transfer energy over long distances through space. This energy transfer happens without the motion or contact of objects.

Light Transfers Energy

Light transfers energy from a light source, such as a light bulb, to your eyes. The energy of light causes changes in your eyes. These changes allow you to see things around you.

The sun is another light source. Light from the sun transfers across 150 million kilometers of space to reach Earth. That's 93 million miles that the sun's light travels to reach us here on Earth! You can see the sunlight reflecting off your skin. Too much of the sunlight can cause an unpleasant change—sunburn! The sunburn is a change, evidence that energy was involved.



Light energy transfers from the sun and causes a change. And a sunburn can be a very unpleasant change.

Sound Transfers Energy

Sound energy often comes from objects that are not touching you, but it causes changes you can hear and sometimes feel. Sometimes when a horn honks nearby or a firework goes off, the sound energy pushes the air, and you feel a push from the sound. Such changes are evidence that an energy transfer happened.

Heat Transfers Energy

Energy also transfers from place to place as heat. If you pour hot tea into a cool cup, soon the cup gets hot. Let a spoon rest in the tea, and soon the spoon gets hot. This is because heat energy transfers from the tea to the cup and to the spoon. The heat energy causes a change. It changes the temperature of the cup and spoon.

The cup and spoon becoming hot is evidence that energy has transferred (moved), because the heat in one material, the tea, moved and produced changes in other objects, the cup and spoon.



The energy of this hot tea transfers through direct contact between the tea, cup, and spoon. Beware of hot food dishes. They can transfer heat to your hands!

Electricity Transfers Electrical Energy

A light bulb without electricity is just a thing made of glass and metal. But when you add electrical energy to the bulb, what changes? It lights up and becomes a source of light, changing a room from dark to brightly lit. The glowing light is evidence that energy has moved from one place to another. The transfer of electrical energy causes the change from an unlit bulb to a lit bulb.

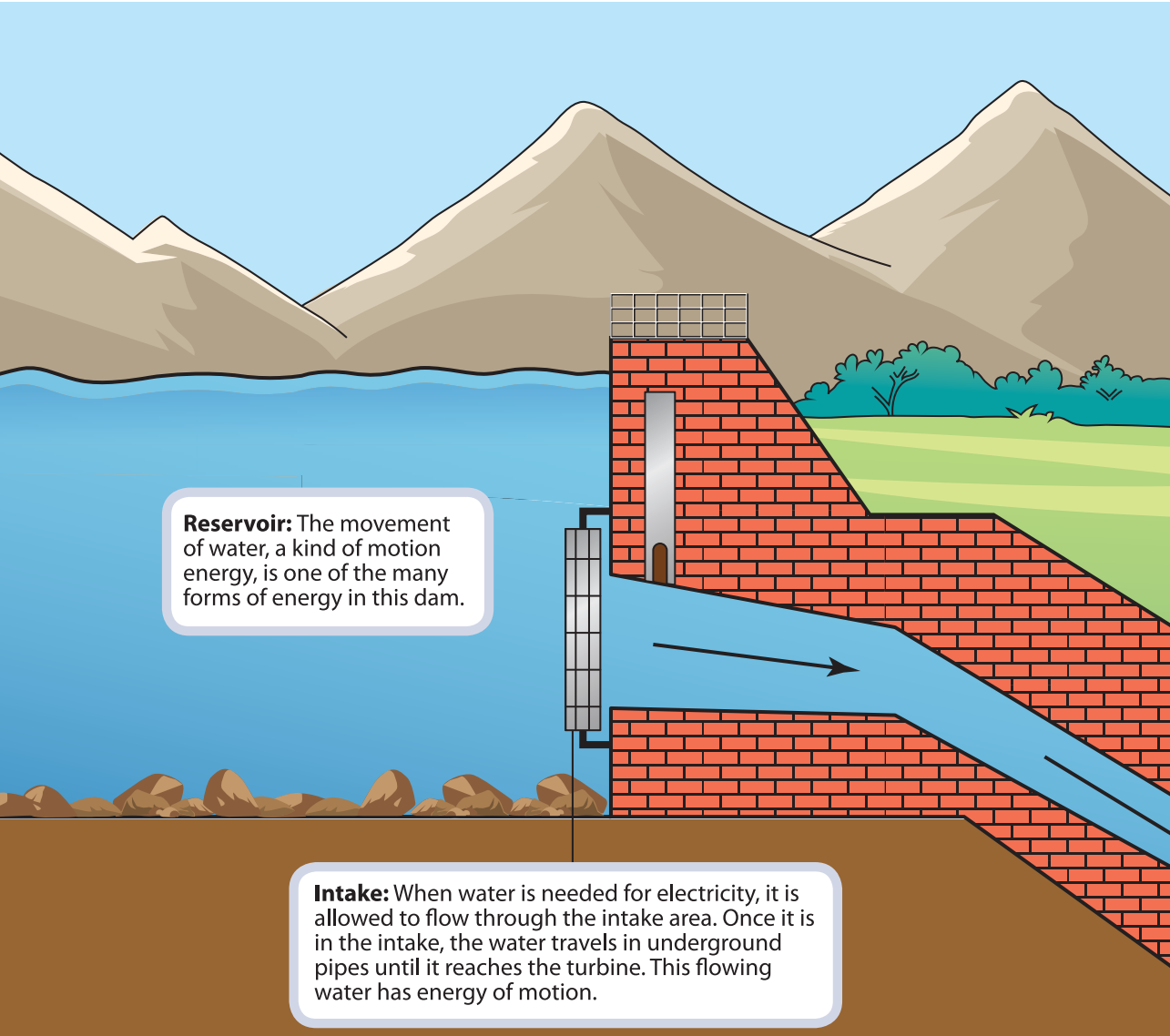
Energy is transferred from place to place by electricity. Like all forms of energy, electrical energy moves from place to place and causes changes. You've probably seen power lines strung between poles along a road. These wires transfer electrical energy from power plants to homes, schools, and businesses. The electrical energy transferred through these wires causes changes when it is used to heat up your toaster or light your room.



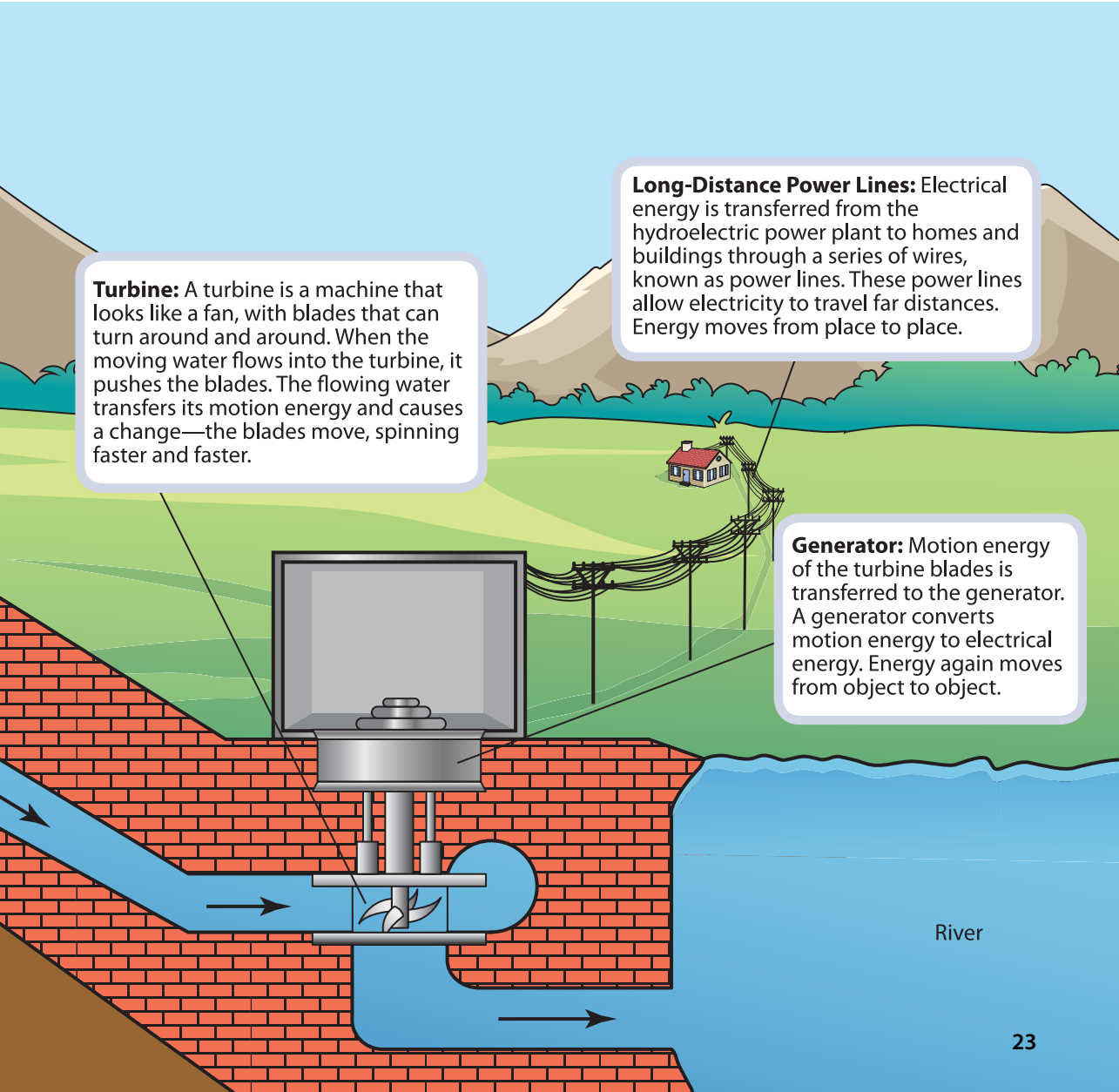
Electrical energy transfers through thick wires like these from power plants to the buildings where people use the electricity.

People Transfer Energy to Improve Society

Engineers often solve problems by making energy readily available and useful. Scientists and engineers work to find new and better ways to use different forms of energy and transfer them from place to place. Hydroelectric power plants use a series of energy transfers so that the motion energy of water can be converted to electrical energy.



Hydroelectric power uses flowing water to move large, spinning turbines. The motion energy of the turbines causes a change that produces electrical energy. And then the energy of electricity moves through wires over long distances, from place to place.



Hoover Dam was built in the Colorado River at the border where Nevada and Arizona meet. You can see in this top photo how the dam holds water back in a great reservoir. When water is allowed to flow through the dam into the lower river, the moving water transfers its energy to equipment inside the dam.





This middle picture of a different dam shows how water moves through a hydroelectric plant with a great deal of energy.

These engineers are building the turbines that go inside a hydroelectric dam. Energy from the moving water transfers to turbines like these. Then the energy transfers in the form of electricity through power lines to buildings where people work and live.

